Diagnosing Small-Model Failures in MemGPT-Style Memory Retrieval

Tool-Call Control, Query Policy, and Teacher-Trajectory Distillation

Hyogi Kim

Ulsan National Institute of Science and Technology

Memory has to outlive the context window

A persistent assistant must answer about things no longer in context:

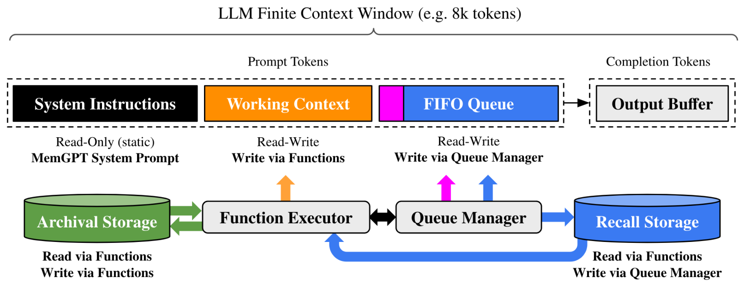
"What artist did I say I could get into?"

The answer lives in a past session—one specific earlier utterance, not world knowledge.

Speaker: "A little bit. I can get into Taylor Swift."

The agent has to recover that utterance and ground the answer in it.

MemGPT: virtual memory for an LLM



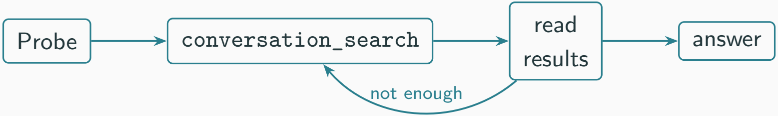
The context window is small, tiered working memory; Archival and Recall storage are the large external memory, reached only through function calls.

In this work we use only the Recall / `conversation_search` path (not archival), and that recall is substring, not semantic.

Figure: Packer et al. (2023).

MemGPT makes recall a control problem

The model is a controller over memory tools—it does not answer at once.



The hidden difficulty is not the answer—it is deciding how to operate the memory system.

One question behind a low score

When a small open model is weak at deep memory retrieval, is it because

Knowledge
it cannot form the answer even *with* the evidence?

Behavior
it cannot *operate* the memory loop to get there?

We answer this by taking the loop apart, one component at a time.

Setup

Setup

- Task: MSC Deep Memory Retrieval, in a local Letta/MemGPT loop.
- Models: Llama-3-8B (student, LoRA target), Mistral-7B (frozen replay), GPT-4.1 (teacher & judge).

The twist: a literal substring retriever

A query returns past messages that *contain it as a substring*. `audio studio location` → nothing. `studio`, `California` → hit.

A good query is a literal phrase likely to appear in past dialogue, not a semantic summary.

How we test it: a diagnostic ladder

Each rung removes or repairs *one* part of the loop, so the next failure reads narrowly.

Intervention	**What it isolates**
Raw vanilla	can it enter the loop at all?
Strict template	just a formatting barrier?
Full history	can it answer without searching?
Teacher-trace replay	can it answer with the right evidence?
LoRA + query hint	does better query selection help?
Skeleton + reranking	can query policy be isolated and fixed?

Findings

They fail before retrieval even starts

- Raw small models: 0 / 5 loops — they never satisfy the tool-call contract.
- A strict formatting adapter restores the loop: 481 / 500 completed…
- … yet containment is only 0.1954.

Fixing the *format* is not fixing the retrieval.

Remove the search, and it jumps

Put all past sessions directly in context — no tool calls, no retrieval:

$$\text{containment} \quad 0.1954 \quad \longrightarrow \quad 0.5080$$

Most of the early loss is the model driving retrieval, not the model answering.

Our teacher: strong, but not gold

We distill GPT-4.1 traces — but the teacher is imperfect too.

- 398 / 500 traces judged correct…
- … yet only 206 / 500 contain the exact reference.

We keep judge-approved traces and audit them — supervision, *not* gold policy.

Given the evidence, they answer fine

Frozen students, replayed the teacher's retrieved evidence (answer only):

	GPT-4.1 judge
Llama-3-8B	0.73
Mistral-7B	0.88

Mistral fails the tool-call gate entirely, yet answers well from evidence.

Answer generation is not the wall.

The bottleneck is getting evidence, not using it

We give the LoRA student different help, all on the same 398 rows:

What the student is given	GPT-4.1 judge	vs. unaided
Nothing (unaided)	0.58	—
+ teacher query strings	0.63	+0.05
+ teacher evidence	0.87	+0.29

Handing over the evidence recovers the model;
handing over the query strings barely moves it.

Query strings matter — but hints don't reach them

Replayed deterministically, teacher queries retrieve the answer-bearing memory far more often (searched subset):

teacher 0.48 vs. student 0.27 (retrieved-reference)

Where the student goes wrong

Reference: *"I have a cat."*

Teacher queries: `pets`, `cat`, `cats`

Student queries: `pet`, `dogs`, `dogs` $\Rightarrow$ answers *"I have a dog."*

Copying the probe, guessing a wrong prior, missing the discriminative literal — good queries exist, the model just doesn't pick them.

Memorizing queries fails; search-time control works

Hard-set SFT / DPO

fix a few hard rows,
hurt the full distribution.
No general query policy.

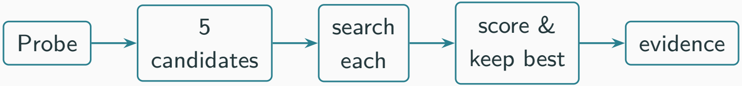
Candidate reranking

propose several queries,
keep the best by local feedback.
25/72 hard cases recovered; first to lift the full set.

Use the retriever as feedback at search time — don't bake teacher queries into weights.

Candidate reranking, step by step

At each search step, let the model guess several times and judge by what comes back:



- The score never sees the answer: it rewards results near a target count and on-topic with the probe, and penalizes *empty*, *too broad*, and *repeated* queries.
- The hard failures were simply zero-result queries — proposing several and dropping the empties already recovers $0/72 \to 25/72$.

Takeaways

Takeaways

- A small memory agent fails in stages: contract $\rightarrow$ retrieval $\rightarrow$ grounding.
- Tool-call control, query policy, and evidence grounding are different problems — one teacher trajectory copies all three too coarsely.
- Build and evaluate small memory agents component by component.
- Put query-policy diagnostics before broad claims about long-term memory.

Thank you

jeff4380@unist.ac.kr